NONPARAMETRIC DENSITY ESTIMATION: A COMPARATIVE STUDY

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Abstract. Motivated by finance applications, the objective of this paper is to assess the performance of several important methods for univariate density estimation focusing on the robustness of the methods to heavy tailed target densities. We consider four approaches: a fixed bandwidth kernel estimator, an adaptive bandwidth kernel estimator, the Hermite series (SNP) estimator of Gallant and Nychka, and the logspline estimator of Kooperberg and Stone. We conclude that the logspline and adaptive kernel methods are superior for fitting heavy tailed densities. Evaluation of the convergence rates of the SNP estimator for the family of Student-t densities reveals poor performance, measured by Hellinger error. In contrast, the logspline estimator exhibits good convergence independent of the tail behavior of the target density. These findings are confirmed in a small Monte-Carlo experiment.

1. Introduction

The primary objective of this paper is to assess the performance of several univariate density estimators focusing on target densities with heavy tails. Motivated by finance applications, it is particularly important to find methods that perform well for heavy-tailed densities. We consider four approaches: a fixed bandwidth kernel estimator of Sheather and Jones (1991), the adaptive bandwidth kernel estimator of Silverman (1986), the Hermite Series (SNP) method of Gallant and Nychka (1987), and the logspline approach of Kooperberg and Stone (1991, 1992).

We argue that the logspline and adaptive kernel methods are superior to the other methods for target densities with heavy tails. In particular, the performance of the SNP estimator relative to the logspline estimator becomes significantly worse as the tail of the density becomes heavier. For algebraically tailed densities like the Student t family the Hellinger error of the logspline estimator with parametric dimension p decreases exponentially in p independent of tail heaviness. On the other hand, the Hellinger error of the SNP estimator decreases algebraically, like $p^{-\nu/2}$, where ν denotes the degrees of freedom of the t-distribution and reflects tail heaviness. These findings are confirmed in a small Monte-Carlo experiment designed to compare the four candidate estimators.

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2. The Methods

Given a sample $\{X_1, X_2, \ldots, X_n\}$, the classical kernel density estimator with Gaussian kernel has the form

$$\hat{f}_{FK}(x) = \frac{1}{nh} \sum_{i=1}^{n} \phi\left(\frac{x - X_i}{h}\right), \tag{1}$$

where ϕ denotes the standard Gaussian density, and a fixed bandwidth h controls the degree of smoothing. In selecting h, we use the two-stage direct plug-in bandwidth selection method of Sheather and Jones (1991), which has been shown to perform quite well for many density types by Park and Turlach (1992) and Wand and Jones (1995).

The adaptive kernel density estimator of Silverman (1986) is defined as

$$\hat{f}_{AK}(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h\lambda_i} \phi\left(\frac{x - X_i}{h\lambda_i}\right), \tag{2}$$

where λ_i is a local bandwidth factor which makes the bandwidth $\lambda_i h$ to narrower near modes and wider in the tails. Based on the pilot estimate by fixed kernel method, $\tilde{f}(x)$, λ_i is determined as $\lambda_i = \tilde{f}(X_i)/g^{-\alpha}$, where $\log g = n^{-1} \sum \log \tilde{f}(X_i)$ with $\alpha = 1/2$. The bandwidth of the pilot estimate is determined by Silverman's (1986) rule-of-thumb, $h = 0.9An^{-1/5}$ where $A = \min\{\text{standard deviation, interquartile range}\}/1.34$.

The Gallant and Nychka (1987) SNP approach is a truncated Hermite series estimator defined as

$$\hat{f}_{SNP}(x, \theta) = \left[\sum_{i=0}^{K} \theta_i w_i(x)\right]^2 + \varepsilon_0 \phi(x), \quad \theta = (\theta_0, \ldots, \theta_K), \quad \sum_{i=0}^{K} \theta_i^2 + \varepsilon_0 = 1, \tag{3}$$

where $w_i(x)$ is a normalized Hermite polynomial, $\phi(x)$ is the standard normal density, and ε_0 is a (small) positive number. Estimation is by quasi-maximum likelihood:

$$\hat{\theta} = \text{argmax}_{\theta,\mu,\sigma} \frac{1}{\mathrm{n}} \sum_{\mathrm{j}=1}^{\mathrm{n}} \log\left\{\frac{1}{\sigma} \hat{f}_{SNP}\left[\frac{X_j - \mu}{\sigma}\right]\right\}, \tag{4}$$

where μ and σ correspond to location and scale. The truncation point is chosen as either $K = n^{1/5}$ or according to the AIC or BIC model selection criterion. The parametric dimension is $p = K + 3$, due to inclusion of μ, σ, and the θ_0 term.

The logspline method of Kooperberg and Stone (1991, 1992) models a log-density function as a cubic spline:

$$\hat{f}_{LS}(x, \theta) = \exp\left\{\sum_{i=1}^{p} \theta_i B_i(x) - c(\theta)\right\}, \; x \in \mathcal{R}, \tag{5}$$

where $c(\theta)$ is a normalization factor, and the functions $\{B_i(x)\}$ denote the standard cubic B-splines. The coefficients θ are estimated by maximum likelihood; knot selection determining the choice of the B-spline basis is based on the BIC criterion.

3. SNP Convergence Rates for Heavy-Tailed Densities

In this section we consider the SNP (Hermite) approximation for the family of Student-t densities and derive the rate of convergence of the SNP approximation, as a function of the dimension of SNP model and the degrees of freedom, ν, of the t-density.

3.1 SNP coefficients and Hellinger error

Under certain regularity conditions, a density f can be represented as an infinite sum of orthonormal Hermite polynomials

$$f(x,\theta) = \left[\sum_{i=0}^{\infty} \theta_i w_i(x)\right]^2, \quad \theta = (\theta_0, \theta_1, \ldots\ldots), \quad \sum_{i=0}^{\infty} \theta_i^2 = 1. \tag{6}$$

The coefficients, θ_i, minimizing Hellinger error can be determined by calculating

$$\theta_i = \int_{-\infty}^{\infty} \sqrt{f(x,\theta)}\; w_i(x)dx. \tag{7}$$

Let $T_K(x,\lambda)$ be a partial sum of normalized Hermite polynomials

$$\sqrt{T_K(x,\lambda)} = \sum_{i=0}^{K} \lambda_i w_i(x), \tag{8}$$

where λ_i are arbitrary. The Hellinger error in approximating $f(x)$ by $T_K(x,\lambda)$, using the othogonality of the w_i is given by

$$H(f,T_K) = \int_{-\infty}^{\infty} (\sqrt{f(x)} - \sqrt{T_K(x,\lambda)})^2 dx = 1 + \sum_{i=0}^{K} (\lambda_i - \theta_i)^2 - \sum_{i=0}^{K} \theta_i^2.$$

Thus, $H(f,T_K)$ is minimized over the λ_i's by choosing $\lambda_i = \theta_i : i = 1, \ldots, K$. Having chosen the λ_i in this way we have

$$H(f,T_K) = \sum_{i=K+1}^{\infty} \theta_i^2, \tag{10}$$

so ε_0 in (3) may be interpreted as the Hellinger error.

3.2 Evaluating the approximation error of the SNP estimator.

Let $p = K + 3$ be the number of parameters. We have computed θ_p up to $p = 200$ using (7) for several densities with varying tail behavior: Student-t distributions with degrees of freedom $\nu = 1, 2, 3, 5, 11$, and a bimodal mixture of normal densities.[1] In Figure 1, we plot for each of the densities, the logarithm of $|\theta_p|$ as a function of p on log scale. The plotted curves are asymptotically linear for the Student-t densities, but curved downward for the bimodal density. Moreover, as the degrees of freedom ν decreases, the plotted slope becomes flatter, indicating that the rate of approximation error improvement measured by Hellinger distance becomes slower as the tail of the t-density becomes heavier.

[1] A mixture of normals such that $\frac{1}{2}N[-1, 0.5^2] + \frac{1}{2}N[1, 0.5^2]$.

This suggests that for algebraic tailed densities we have the relation,

$$\theta_p = Ap^{\alpha}. \tag{11}$$

Supposing (11) holds, consider approximating the sum $H(f, T_K) = \sum_{i=K+1}^{\infty} \theta_i^2$ by the integral $h(p) = \int_p^{\infty} \theta^2(v)dv$. For (11) we have,

$$h(p) = \int_p^{\infty} \theta^2(v)dv = A^2 \int_p^{\infty} v^{-2\alpha}dv = A^2(2\alpha - 1)p^{-(2\alpha-1)}. \tag{12}$$

The value for α corresponds to the slope of the Student-t lines in Figure 1. Least squares estimates of α yield

$$\alpha = (\nu + 2)/4, \tag{13}$$

over the range of ν studied. Based on (13), we conclude that Hellinger error decreases like,

$$h(p) \sim p^{-\nu/2}. \tag{14}$$

If $p = n^{\gamma}$ is assumed, the Hellinger rate of convergence becomes $h(p) \sim n^{-\nu\gamma/2}$.

Recently, Coppejans and Gallant (2000) have shown that for $p = n^{\gamma}$ the best Hellinger error rate for the SNP density estimator is $h(p) \sim n^{-k\gamma/2}$, where k is determined by the conditions in the following lemma.

Lemma (Fenton and Gallant, 1996b) *Let f be a density such that $f(x) = g^2(x)e^{-x^2/2} + \epsilon_0\phi(x)$. If g is k-times differentiable with $g^{(j)} \in L_2(e^{-x^2/2})$ for $j = 0, 1, \ldots, k$, then the Hermite coefficients of g satisfy $\sum_{i=p}^{\infty} \theta_i^2 = o(p^{-k})$.*

This result seems to suggest that provided f is sufficiently smooth, as determined by the order of the differentiability, k, the rate of convergence of the SNP approximation is good. While the differentiability of g in the foregoing lemma may be considered quite innocuous, the requirement that that $g^{(j)} \in L_2(e^{-x^2/2})$, turns out to be surprisingly restrictive for the class of heavy tailed f. Recall that $g^{(j)} \in L_2(e^{-x^2/2})$ iff $\int_{-\infty}^{\infty} (g^{(j)}(x))^2 \, e^{-x^2/2}dx < \infty$.

For the family of Student-t densities one can verify by straightforward computations that this condition fails for $k > \nu/2$ for even ν and for $k > (\nu-1)/2$ for odd ν. This is easily seen by computing $\lim_{x\to\infty} g^{(j)}(x)$ and noting that the integral diverges unless the limit tends to zero. This reconciles the Coppejans and Gallant result with our findings showing that the convergence of the SNP estimator is highly sensitive to the tail behavior of the target density, with poor performance at heavy tailed densities like the Student-t with low degrees of freedom.

4. Logspline Convergence Rates for Heavy-Tailed Densities

To contrast the SNP results in the previous section with the behavior of the logspline estimator we have computed the Hellinger error of the p dimensional logspline estimator based on artificial observations at equally spaced quantiles of the target density. Modifying the logspline routines of Kooperberg and Stone (1991, 1992), we

replace the summation in (5) by integration so that the true density can be approximated, and the effect of knot placement on the rate of convergence is kept to a minimum. The Hellinger error of the logspline method is found to decrease faster than p^{-7}, and suggesting an exponential law of the form $e^{-\alpha p^{\tilde{\beta}}}$, with parameters independent of tail behavior. For algebraic tailed densities like the Student-t family this represents a considerable gain over the SNP method.

Figure 2 illustrates our comparison for the case of the $t(3)$ density. The solid line is the Hellinger error for the SNP method as a function of the number of parameters p as developed in Section 3. The dotted line is the counterpart for the logspline method. The open and filled circles indicate the Monte Carlo simulation results for the Hellinger error of the logspline method and the SNP method, respectively. It is seen that the Hellinger approximation error of the logspline method decreases significantly faster than that of the SNP method, and the rate is consistent with the Monte Carlo simulation results.

5. Monte Carlo Simulation Study

This section compares the performance of our four density estimators in fitting heavy-tailed densities in a small Monte Carlo study. In addition to the normal density, $t(3)$ and $t(1)$ densities are chosen to see the effect of tail behavior. The lognormal density is included as different type of heavy tail. All the densities are rescaled so that 99.9% of the mass is included in the range of $[-3, 3]$. We evaluate the performance by the Hellinger error defined as

$$H_M(\hat{f}, f) = \int_{-M}^{M} (\sqrt{\hat{f}(x)} - \sqrt{f(x)})^2 dx. \tag{15}$$

The Hellinger criterion accentuates the error in the tails. We set $M = 3$.

The source code used for the Monte-Carlo study for the SNP estimator is that of Fenton and Gallant (1996a), for the logspline method that of Kooperberg and Stone (1991, 1992). The adaptive kernel estimator is based on Silverman (1986). The direct plug-in fixed kernel method is coded by Wand and Jones (1995). In the Monte Carlo, the default settings are used for all the algorithms. The logspline program includes knot deletion procedure. The numbers of parameters p for the SNP and logspline estimators are determined minimize the BIC criterion.[2]

Table I summarizes the Hellinger performance for each of the densities studied. The sample size is fixed at 500; the error statistics are the average of 500 replications. The errors of the logspline and adaptive kernel methods are quite similar for all the densities. Those of the SNP and fixed kernel methods increase as the tails become heavier. The performance ranking for heavy tailed densities is 1) logspline, 2) adaptive kernel, 3) fixed kernel, and 4) SNP. In particular, the performance of the SNP estimator relative to that of the logspline estimator becomes significantly worse as the tail of the density becomes heavier.

[2]The upper bound of the Hermite polynomial order K is set as $K = n^{1/2}$ to avoid problems of non-convergence of the algorithm.

To assess the qualitative performance in fitting heavy-tailed densities, the $t(3)$ density is estimated using each of the four methods for one realization of a random sample, and the results are illustrated in Figure 3. The dotted lines indicate the true $t(3)$ density while the solid lines denote the estimates. The SNP estimate exhibits large spurious oscillations in the tails, which may be attributed to the effect of the higher order terms in the Hermite expansion. The fixed kernel estimate is also problematic especially in the tails.

6. Empirical Example

In Figure 4 we compare density estimates for the daily change of $S\&P500$ index. The daily price change is $\Delta p = \log p_t - \log p_{t-1}$. In this figure we observe the same pattern of differences in the tail shape of the estimates exhibited in the previous figure, suggesting that the lobes in the tails of the fixed kernel and SNP estimates are likely to be artifacts of the fitting method rather than real features of the data.

This observation is relevant to the ongoing dispute over the existence of side lobes in the tails of densities of asset price changes. Based on SNP density estimation of daily pound/dollar exchange rate data from 1974 to 1983, Gallant, Hsieh, and Tauchen (1991) concluded that side lobes existed in the error density of their ARCH specification. Engle and Gonzalez-Rivera (1991) reported similar lobes in the distribution of the standardized residuals from a GARCH(1,1) model. They suggested that the detected side lobes reflect the residual non-linearity which could not be explained by the ARCH/GARCH formulation. In contrast, studies by Bolleslev (1987) and Baillie and Bollerslev (1989) analyzing data on the British pound for 1980-1985, found little evidence against GARCH(1,1) with t-distributed errors. Bollerslev, Chou, and Kroner (1992) suggested that tail oscillations were unduly influenced by a few outliers rather than a feature of the entire sample period. This conclusion seems broadly consistent with the findings reported above.

7. Conclusion

It is clear that for heavy-tailed densities like Student-t, the rate of convergence of the SNP estimator is remarkably slow compared to logspline or even conventional kernel methods. Since financial data are often characterized as heavy-tailed, it seems prudent to consider the logspline methods or adaptive kernel method for financial applications.

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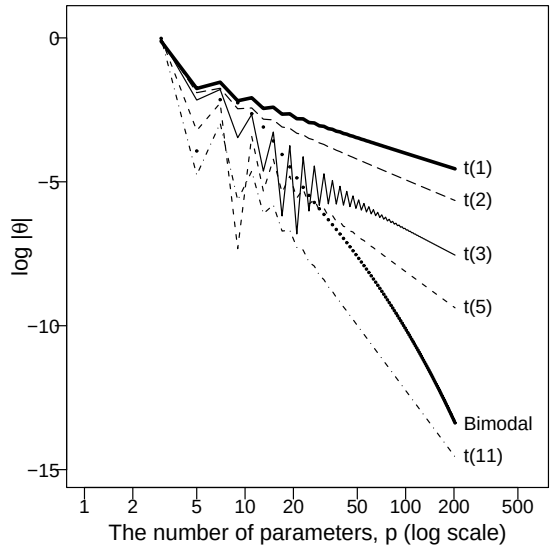

FIGURE 1. The coefficients of Hermite polynomial terms of the SNP estimates as a function of the number of parameters in approximating true densities.

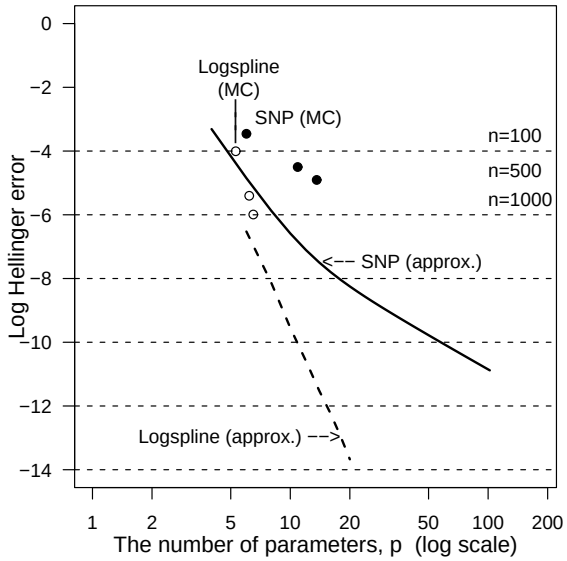

FIGURE 2. The Hellinger error as a function of the number of parameters. Points labelled as "MC" indicate the results from the Monte Carlo study. Open circles represent the Monte-Carlo performance of the logspline estimator, filled circles the performance of the SNP estimator. Lines labelled as "approx." indicate the approximation error in fitting true densities.

TABLE I. Simulation Results for Densities with Heavy Tails

(Sample size 500, average of 500 simulated samples)

	Method	**Normal** mean	(s.d.)	**t(3)** mean	(s.d.)	**t(1)** mean	(s.d.)	**Lognormal** mean	(s.d.)
Hellinger	F.Kernel	0.0035	(0.0017)	0.0112	(0.0026)	0.1344	(0.0061)	0.0237	(0.0032)
	A.Kernel	0.0035	(0.0016)	0.0049	(0.0020)	0.0247	(0.0038)	0.0121	(0.0022)
	SNP	0.0010	(0.0010)	0.0111	(0.0069)	0.4242	(0.2946)	0.0383	(0.0168)
	Logspline	0.0029	(0.0016)	0.0045	(0.0026)	0.0087	(0.0038)	0.0052	(0.0025)
p	SNP	3.0	(0.0)	10.9	(4.2)	25.3	(1.5)	22.9	(2.9)
	Logspline	4.2	(0.6)	6.2	(0.5)	9.4	(0.8)	5.8	(1.0)

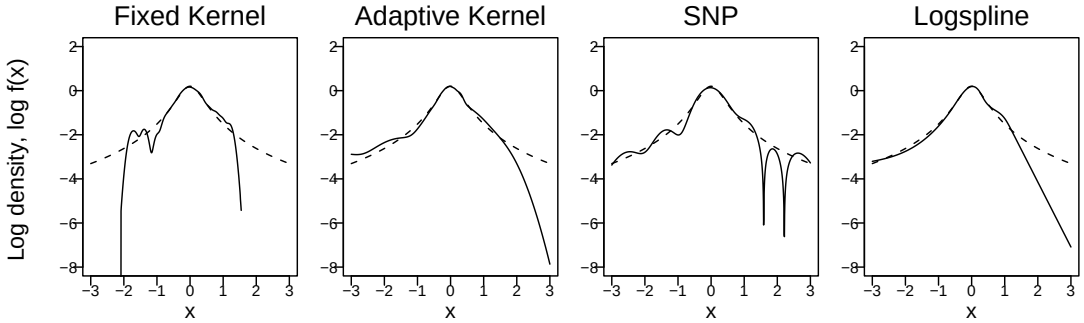

FIGURE 3. Estimates of $t(3)$ log densities. Dotted lines indicate the true $t(3)$ density; solid lines denote the estimates.

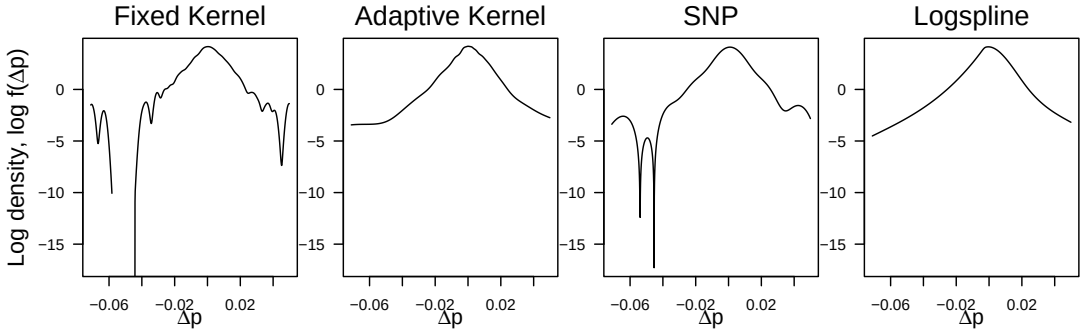

FIGURE 4. Four density estimates for daily changes in $S\&P500$ index. Sample period is 1989-1998. Note the similarity to the prior figure based on artificial t(3) data.